



FIRST SEMESTER 2026-27

Course Handout Part II

Date: 03/08/2026

In addition to part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : MATH F212
Course Title : OPTIMIZATION
Instructor-in-charge : SHIVI AGARWAL (shivi@pilani.bits-pilani.ac.in; 6067-T)
Instructor : Shivi Agarwal (shivi@pilani.bits-pilani.ac.in; 6067-T),
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1. **Course description:** Introduction to optimization; linear programming; simplex methods; duality and sensitivity analysis; transportation model and its variants; integer linear programming nonlinear programming; multi-objective optimization; evolutionary computation techniques.

2. **Scope and Objective:** Engineers, scientists, analysts, and managers often face the challenge of making trade-offs between different factors to achieve desirable outcomes. Optimization is the process of making these trade-offs in the best possible way. Over the years, optimization problems have become well-established and are encountered in the physical sciences, engineering, economics, industry, planning, and many other areas of human activity. The objective of this course is to familiarize students with standard methods of solving optimization problems.

This course covers various aspects of optimization, including a description of techniques such as Linear Programming and Nonlinear Programming, and their applications in various engineering and science disciplines, including economics and finance. Multi-objective optimization, which addresses optimization with more than one objective, is also discussed. Additionally, the course provides a brief and informative overview of the non- traditional optimization technique, Genetic Algorithms.

3. **Course learning outcomes:**

CLO1: Understanding of the formulation and solution of LPP

CLO2: Understanding of the duality theory and post optimal analysis with their applications in transportation, assignment and integer LPPs

CLO3: Understanding of multi-objective LPP, Nonlinear programming and non-traditional optimization techniques





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4. Text Book:

[TB1] H A Taha, *Operations Research: An Introduction*, Pearson Education, 9th Edition, 2011.

Reference Books:

[RB1] J C Pant, *Introduction to Optimization: Operations Research*, Jain Brothers, New, 6th Edition, 2004.

[RB2] WL Winston, *Operations Research: Applications and Algorithms*, Thomson Learning, 4th Edition, 2004

[RB3] A Ravindran, DT Philips and JJ Solberg, *Operations Research: Principles and Practice*, John Wiley & Sons, Singapore, 2nd Edition, 1987



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5. Course Plan:

Wk no.	Module no.	Description	Lec. No.	Ref. [TB1]	Learning Outcomes
1	1	Introduction to Optimization	1		Motivation to study optimization
1-2	2	Two Variable LP Model, Graphical LP Solution, LP Problems, Convex Set	2-6	(2.1, 2.2, 7.1, 7.1.1)	Develop and solve Linear Programming (LP) models using the graphical solution method
3-5	3	LP Model in Equation Form, Transition from Graphical to Algebraic Solution The Simplex Method, Generalized simplex Tableau in Matrix Form, Revised Simplex Method, Artificial Starting Solution, Special cases in the Simplex Method	7-13	(3.1,3.2, 3.3., 7.1.2, 7.2, 3.4, 3.5)	Understand the rationale and process of simplex method and discuss special cases in Linear Programming
5-7	4	Definition of Dual Problem, Duality, Primal-Dual Relationships, Additional Simplex Algorithms (Dual Simplex Method, Generalized Simplex Algorithm), Post Optimal Analysis	14 -19	(4.1, 7.4,4.2, 4.4, 4.5)	Understand duality concepts, interpret dual solutions, relate them to primal solutions, and use of post-optimal analysis for decision making
7-8	5	Definition of Transportation Problem, The Transportation	20-22, 23-24	(5.1, 5.3, 5.4)	Formulation and solution of transportation and assignment problems





		Algorithm, The Assignment Model			as LPP
9	6	Goal Programming Formulation, Algorithms: Weights Method	25-27	(8.1, 8.2)	Understand multiples objectives optimization and how to solve multi objective optimization problems
10	7	Formulation of IP Problem Branch and Bound Method for Solving IPP	28- 30	(9.1, 9.2.1)	Formulation and solution of Integer Programming problem
11-13	8	Unconstrained Problems, Convex and Concave Functions, Elimination Methods: Direct Search Method Gradient of a Function, Descent Methods: Steepest Descent Method, Karush-Kuhn-Tucker (KKT) Conditions, Quadratic Programming	31, 32- 35, 36-38	(20.1, 20.1.1) (21.1.1, 21.1.2) (20.2.2) (21.2.2)	Solution of Nonlinear Programming problem
13-14	9	Drawbacks of the Classical Techniques, Introduction to Nontraditional Optimization Technique (Genetic Algorithms)	39- 40	Class Notes	Understanding of Nontraditional Optimization technique



6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Tests (best 2 out of 3)		20	TBA Announced	CB
Mid Semester	90 minutes	35	TBA by AUGSD	CB
Comprehensive	180 minutes	45	TBA by AUGSD	(OB/CB)

7. **Criterion for NC:** The student with total marks below **10%** of the course total will be awarded NC grade.
8. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation.
9. **Make-up Policy:** Mid semester and comprehensive examination make-ups will be granted only to genuine cases. For cases related to illness, prior information and proper documentary evidence are essential. **No make-up for the tests.**
10. **Problems:** Students are strongly advised to work out all the problems in the text-book and do similar problems from the reference books. It is also strongly recommended that the students should try out the algorithms on computers to get a better understanding of the subject.
11. **Chamber Consultation Hours:** To be announced in the class.
12. **Notice:** Notices concerning this course will be displayed on NALANDA.

**INSTRUCTOR-IN-CHARGE
(MATH F212)**

